



A Report on
Workshop on
Programmable Logic Controllers: Hands-On
In association with
Birla Vishvakarma Mahavidyalaya, V.V.Nagar
Organised By
THE ROBOTICS SOCIETY STUDENT CHAPTER
25-10-2023



B.V.M Engineering College [An Autonomous Institution]

Managed By Charutar Vidya Mandal

TRS students chapter organized

Workshop on

Programmable Logic Controllers: Hands-On

Venue: IDEA Lab, AD Patel Institute of Technology, Vallabh Vidyanagar, Anand

The PLC Learning and Hands-on workshop held at IDEA Lab, AD Patel Institute of Technology in Vallabh Vidyanagar, Anand, was an enriching experience for all participants. The workshop was conducted by seasoned instructors, Mr. Dhaval Tailor and Mr. YD Patel, both of whom brought their extensive knowledge and expertise to the event. The main agenda of the workshop was to provide participants with practical insights into the use of Programmable Logic Controllers (PLCs) in real-life applications. This report will delve into the workshop's vision, outcomes, content, and the focus on logical gates and PLCs.

Vision:

The overarching vision of this workshop was to equip participants with a hands-on understanding of PLCs and how they are used in various industrial applications. The organizers aimed to bridge the gap between theoretical knowledge and practical skills, empowering attendees to apply PLC technology effectively in real-world scenarios.

Outcomes:

1. **Hands-on Experience of PLC:** Participants gained practical experience in programming and operating PLCs. This hands-on exposure enabled them to apply their knowledge in real-life situations, making them more adept at troubleshooting and optimizing PLC systems.
2. **Exploration of Various Fields Using PLC:** The workshop exposed attendees to a wide range of fields where PLCs are used, from manufacturing and automation to energy management and process control. This exploration broadened their horizons and showcased the versatility of PLC technology.
3. **Understanding Industrial Applications:** Participants left the workshop with a deep understanding of how PLCs are utilized in industrial settings. This knowledge is crucial for those seeking a career in fields where automation and control systems play a pivotal role.

Workshop Structure:

The workshop was structured to provide a comprehensive learning experience. Participants were divided into four teams, each responsible for creating logical gates and demonstrating PLCs on both real and simulated setups.

- Logical Gates: The primary focus of the workshop was on logical gates, including OR, AND, XOR, NOR, and more. Participants were guided through the theoretical aspects of these gates and then tasked with implementing them in practical scenarios using PLCs.
- PLC Application: The instructors provided in-depth knowledge about PLCs, explaining their components, functions, and programming. Participants learned how to program PLCs to control various devices and systems, including motors, sensors, and actuators.
- Real Setup and Simulation: The hands-on approach included working with actual PLC hardware and software simulations. This allowed participants to see the immediate impact of their programming on different systems. The combination of real and simulated setups provided a holistic learning experience.

The workshop's practical and interactive approach ensured that participants not only grasped the theoretical aspects of PLCs but also gained the confidence to apply this knowledge in real-life situations.

Summary:

In conclusion, the PLC Learning and Hands-on workshop at IDEA Lab, AD Patel Institute of Technology, was a resounding success. Participants left with a solid foundation in PLC technology and a clear understanding of its applications in diverse industries. This hands-on experience, guided by expert instructors, undoubtedly empowered attendees to pursue careers in automation, control systems, and related fields with confidence and competence. The workshop played a vital role in achieving its vision of providing participants with a valuable insight into the world of PLCs and their real-world application.

Glimpse of Workshop:



